Hybrid classical-quantum combinatorial optimization algorithms for scheduling in Cell-Free communication systems

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1 Introduction

Cell-free massive multiple-input multiple-output (MIMO) replaces the cellular layout with a large number of distributed access points (APs) that jointly serve the user equipments (UEs) in a coverage area, without cell boundaries Ngo et al. [2017], Interdonato et al. [2019]. Its main promise is uniform service: a UE is never at the edge of a cell, because there are no cells. The price is coordination. If every AP had to serve every UE, the processing and fronthaul load would grow with the size of the network, so practical systems adopt a *user-centric* operation in which each UE is served by a small subset of nearby APs Björnson and Sanguinetti [2020], Demir et al. [2021]. Deciding which antenna serves which UE, and doing so again every time the scenario changes, is therefore part of making cell-free networks scalable.

This decision is a combinatorial problem. In every timeslot, each antenna has a limited number of channels, some UEs cannot share an antenna at all, and UEs that do share one degrade each other through adjacent channel leakage. Selecting the set of UEs an antenna serves is then a weighted selection problem on a conflict graph, closely related to the maximum weighted independent set (MWIS) problem that underlies throughput optimal wireless scheduling Tassiulas and Ephremides [1992]. MWIS is NP-hard on general graphs, and in a scheduling context it must be solved repeatedly, once per slot and per resource, under tight time constraints. Practical schedulers therefore rely on heuristics, most commonly greedy rules, whose solution quality is not guaranteed.

Neutral atom quantum processors offer a natural candidate for this kind of problem. Atoms excited to Rydberg states cannot be simultaneously excited when they are close to each other, an effect known as the Rydberg blockade, so the low energy states of a suitably arranged atom register correspond to independent sets of a graph Pichler et al. [2018], Henriet et al. [2020]. Maximum independent set problems with hundreds of atoms have been addressed experimentally on such platforms Ebadi et al. [2022], and the interactions between

atoms can also be used to encode quadratic unconstrained binary optimization (QUBO) problems whose couplings are non negative Pasqal [a,b]. Hybrid approaches, in which the quantum processor acts as a sampler inside a classical algorithm rather than replacing it, have already been proposed for graph problems on this hardware da Silva Coelho et al. [2023].

This report studies how to bring the antenna assignment problem of a dynamic cell-free network to a neutral atom processor. Instead of building one large graph for the whole schedule, whose size grows with the number of UEs and timeslots, we decompose the problem by antenna: in every timeslot, each antenna solves a small problem over the UEs that are within its range and not yet assigned. Capacity and full coverage are handled classically, so that the problem sent to the quantum side keeps the structure the hardware supports natively. The main contributions are:

- A per antenna, per timeslot formulation of user assignment in a dynamic cell-free scenario, with hard (blockage) and soft conflicts (adjacent channel leakage) encoded as graph edges and node weights that combine link quality and service continuity.
- A QUBO formulation of each per antenna problem whose off-diagonal terms are all non-negative, as required by the neutral atom embedding.
- A complete classical pipeline that serves as a baseline and as the classical frame in which a quantum sampler can be inserted.
- An assessment of how well the resulting QUBOs can be embedded in an atom register, and of which problem structures make the embedding difficult.

The rest of the report is organised as follows. Section 2 reviews related work on user centric cell-free, conflict graph scheduling, quantum optimisation in wireless communications, and neutral atom optimisation.

2 State of the Art

2.1 User centric cell-free

Cell-free massive MIMO was shown to outperform small cell deployments in per-user throughput, in particular for the worst served users, by letting many distributed APs cooperate Ngo et al. [2017]. The original formulation, in which every AP serves every UE, does not scale: the computational complexity and fronthaul signalling grow without bound as the network grows Björnson and Sanguinetti [2020]. Scalable designs restrict each UE to a cluster of APs, which leads to the user centric architecture formalised in Demir et al. [2021], where the central challenge is precisely to retain the benefits of cell-free operation with complexity and fronthaul that remain feasible in very large networks.

How these clusters are formed has received considerable attention. Proposals range from simple rules based on the strongest channels or on the master AP

of each UE Björnson and Sanguinetti [2020] to optimisation based and learning based association. Recent work, for example, trains recurrent neural networks to predict scalable UE–AP associations Di Gennaro et al. [2025]. My previous work studied the same scheduling setting with machine learning techniques Benavente Vilas et al. [2025]. Most of this literature focuses on which APs serve a UE, and treats each AP as able to serve its UEs on the same time–frequency resources. Here we consider instead the finer resource view in which each antenna has a limited number of channels and UEs that share an antenna interact, which turns the assignment into a sequence of small combinatorial problems.

2.2 Scheduling as weighted independent set on conflict graphs

A long line of work models wireless scheduling on a conflict graph, whose vertices are candidate transmissions and whose edges join transmissions that cannot be active simultaneously Jain et al. [2003]. In this model, a feasible schedule is an independent set of the graph. Tassiulas and Ephremides showed that the policy that activates, in every slot, a maximum weighted independent set of links is throughput optimal Tassiulas and Ephremides [1992]. Since MWIS is NP-hard on general graphs, implementing this policy exactly is impractical even with centralised computation, and most deployed or proposed schedulers settle for greedy or maximal independent sets, trading optimality for speed. Node weights are commonly chosen to balance efficiency and fairness, for instance following proportional fairness Kelly et al. [1998].

Our per antenna problems inherit this structure. Hard conflicts are edges that forbid two UEs from sharing an antenna, so in their presence the selected set must be independent. Soft conflicts, such as the rate lost to adjacent channel leakage, are not forbidden but penalised, which turns the MWIS into a QUBO with weighted pairwise penalties. As in classical scheduling, the problem is solved once per resource and per slot, which makes the quality of the per problem solver, and not only its speed, decisive for the final schedule.

2.3 Quantum optimisation in wireless communications

Quantum optimisation has been explored for several wireless problems. An early landmark is QuAMax, which mapped maximum-likelihood MIMO detection to an Ising problem and solved it on a D-Wave 2000Q quantum annealer Kim et al. [2019]. A recent tutorial and survey covers the use of quantum algorithms across wireless resource allocation, scheduling, edge computing and other networking tasks Zhao et al. [2024].

Within cell-free and distributed antenna systems, the closest works address neighbouring problems rather than per antenna user scheduling. Pilot assignment has been reformulated so that it can be solved directly with the quantum approximate optimisation algorithm (QAOA), reaching performance close to exhaustive search in simulation Ma et al. [2026]. Uplink power control under a max–min fairness criterion has been discretised into a QUBO and evaluated

against brute force and greedy baselines with classical and quantum solvers Phan [2025]. User clustering and antenna association in distributed antenna networks have been formulated as a QUBO and solved by quantum annealing Hanakago et al. [2021]. These works rely on gate based QAOA or on annealers with fixed qubit connectivity.

2.4 Neutral atom optimisation and QUBO embedding

On neutral atom processors, the Rydberg blockade forbids two atoms closer than a blockade radius from being excited together, so that the dynamics of the register naturally explore the independent sets of the corresponding *unit-disk* graph Pichler et al. [2018], Henriet et al. [2020]. This property has been used to address maximum independent set problems on unit disk graphs with hundreds of atoms Ebadi et al. [2022], and pulse sequences for this purpose can be designed and emulated.

The restriction to unit disk graphs is the main obstacle for applications: deciding whether a graph admits a unit disk realisation is itself NP-hard Breu and Kirkpatrick [1998], and many graphs, such as large stars, admit none. Two strategies exist. The first uses gadgets: Nguyen *et al.* construct explicit mappings from MWIS on graphs of arbitrary connectivity, and from general QUBO problems, to MWIS on unit disk graphs with at most a quadratic overhead in the number of qubits Nguyen et al. [2023]. Cazals *et al.* prove that such polynomial reductions necessarily incur this quadratic blow up and degrade approximation guarantees, and propose instead a divide and conquer heuristic with linear overhead that works on precalibrated atom layouts Cazals et al. [2025]. The second strategy embeds the problem directly: atom positions are chosen so that the pairwise interactions, which decay as the inverse sixth power of the distance, reproduce the off-diagonal coefficients of the QUBO, while the diagonal terms are applied as detunings Pasqal [a]. Because the interaction is always repulsive, this only applies to QUBOs whose off-diagonal terms are non negative, a restriction that Pasqal's `qubo-solver` library also imposes Pasqal [b]. Force directed graph layouts have been used to build such registers when no exact realisation is available da Silva Coelho et al. [2022]. General background on QUBO and Ising formulations of combinatorial problems can be found in Lucas [2014], Glover et al. [2022].

Finally, da Silva Coelho *et al.* show how a neutral atom sampler can be embedded in a classical column generation framework, where it generates weighted independent sets for the pricing subproblems of a vertex colouring formulation. In emulation, it needs fewer iterations than classical column generation and reaches better solutions than greedy and random sampling pricing da Silva Coelho et al. [2023]. The same work uses a greedy algorithm that iteratively extracts independent sets as a baseline. This hybrid view, in which the quantum device samples good candidates and the classical algorithm keeps feasibility and global coordination, is the one we adopt.

2.5 Positioning of this work

The works above leave a gap at their intersection. Quantum approaches in cell-free MIMO address pilot assignment, power control or clustering, mostly with QAOA or quantum annealers, while neutral atom optimisation has been demonstrated mainly on generic graph benchmarks and on problems from other domains. To the best of our knowledge, per antenna user scheduling in a dynamic cell-free network has not been formulated as a QUBO tailored to neutral atom hardware. Our formulation keeps each subproblem small enough for current emulators, keeps all couplings non negative so that direct embedding is possible, and leaves the constraints that would break this structure (capacity and coverage) to the classical part of the pipeline, in line with the evidence that general reductions to unit disk graphs are costly Cazals et al. [2025].

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